

Determine the dimension of, and a basis for, the row space of the matrix

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix}$$

Solⁿ: Reduce it to row echelon form

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{pmatrix} 1 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix} \begin{matrix} R_2' = R_2 - 2R_1 \\ R_4' = R_4 - R_1 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_2' = -R_2}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix} \begin{matrix} R_3' = R_3 - 2R_2 \\ R_4' = R_4 - R_2 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow{R_4' = R_4 - 2R_3}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So, Basis for $R(A)$ is $\{(1, 0, 1), (0, 1, -1), (0, 0, 1)\}$

So, $\dim R(A) = 3$

rank of A is $= 3$

Basis for $C(A)$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$

$\dim C(A) = 3$
rank $(A) = 3$

EX Find a basis for the subspace of $\mathbb{R}^4$ spanned by the set of vectors $\{(1, 1, -4, -3), (2, 0, 2, -2), (2, -1, 3, 2)\}$.

Solⁿ: Form a matrix whose rows are given vectors

$$\begin{bmatrix} 1 & 1 & -4 & -3 \\ 2 & 0 & 2 & -2 \\ 2 & -1 & 3 & 2 \end{bmatrix} \begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 2R_1 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -4 & -3 \\ 0 & -2 & 10 & 4 \\ 0 & -3 & 11 & 8 \end{bmatrix} \begin{array}{l} R_2' = -\frac{1}{2}R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -4 & -3 \\ 0 & 1 & -5 & -2 \\ 0 & -3 & 11 & 8 \end{bmatrix} \begin{array}{l} R_3' = R_3 + 3R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -4 & -3 \\ 0 & 1 & -5 & -2 \\ 0 & 0 & -4 & 2 \end{bmatrix} \begin{array}{l} R_3' = -\frac{1}{4}R_3 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -4 & -3 \\ 0 & 1 & -5 & -2 \\ 0 & 0 & 1 & -\frac{1}{2} \end{bmatrix}$$

which is in row echelon form.

$$\therefore \text{Basis } \left\{ (1, 1, -4, -3), (0, 1, -5, -2), (0, 0, 1, -\frac{1}{2}) \right\}$$

Dimension of the subspace is 3.